

A SHIFT IN SOFTWARE ABSTRACTION

High-level languages

Fortran, C

OS abstractions

processes, virtual memory

ML workloads

deep learning

Foundation models

Software 3.0: prompt a model

each shift pulls an architecture response

THE ARCHITECTURE RESPONSE

Optimizing compilers

and ISA design

Processor features

protection, privilege levels

Specialized accelerators

GPU, TPU

Architecture 2.0

the design loop itself

changes WHAT architects build

changes HOW architects build

Prior waves changed the artifact; foundation models change the design loop.